

BSA's Contractual Representative	Overall Value Amount	Contract Type	Contract Number
Mark Brock	TBD	Firm Fixed Price	476517

BROOKHAVEN NATIONAL LABORATORY

Brookhaven Science Associates

P.O. Box 5000

UPTON, L.I., N.Y. 11973-5000

Name and Address of Contractor

CONTRACT

This Contract (the "Contract") is between the party named above (the "Contractor"), and Brookhaven Science Associates, LLC ("BSA"), the latter acting under a Prime Contract (DE-SC0012704) with the United States of America (the "Government") represented by the United States Department of Energy ("DOE").

I. SCOPE OF WORK

The Contractor shall provide one **Dilution Refrigerator (Ultralow Temperature Quantum Materials/Devices measurement system (UTQMD))**.

All work shall be conducted in strict accordance with the attached BSA, Statement of Work, dated 2/27/2026, and Specification No. SPC-UTQMD-CFN-001, dated 02/05/2026.

- A. Reports/Deliverables:** The Contractor shall provide the following deliverable(s) to BSA's Technical Representative (Tech Rep):

Item	Deliverable	Due
1	Program Plan	2nd week after award of contract
2	Technical and Progress Teleconference	2nd week after award of contract/monthly
3	Performance Reports	5th of each month after receipt of Program Plan and within 2 weeks of key project milestones
4	Preliminary Technical Design	4th week after contract award
5	Final Technical Design	5th week after contract award

6	Equipment Footprint and Laboratory Requirements	6th week after contract award
7	Manufacturing/Inspection/Test Plan	6th week after contract award
8	Factory Acceptance Testing	50th week after award of contract
9	Installation Work Plan, JHA	52nd week after award of contract
10	Delivery of the Instrument	52nd week after award of contract
11	Site Acceptance Test and Commissioning Report	56th week after award of contract

II. PERIOD OF PERFORMANCE

This Contract shall be effective as of the date executed by both parties. It shall remain in effect through TBD.

III. FIRM FIXED PRICE AND PAYMENT

- A. **Firm Fixed Price:** In full consideration of the Contractor's performance hereunder, BSA shall pay the Contractor the firm fixed price of TBD U.S. Dollars (\$0.00). The said sum shall constitute full compensation for all services and materials furnished hereunder.
- B. **Payment:** Payment will be made upon receipt and approval of properly certified invoices. Payment will be in accordance with the following milestone payment schedule:

Milestone	Amount
BSA receipt and acceptance of Preliminary Technical Design Submitted	TBD
BSA receipt and acceptance of Final Technical Design Approved	TBD
BSA receipt and acceptance of Factory Acceptance Test Report	TBD
BSA receipt and acceptance of Installation Work Plan, JHA	TBD
BSA receipt and acceptance of Delivery to BNL	TBD
BSA receipt and acceptance of Commissioning of instrument	TBD
BSA receipt and acceptance of Training on instrument	TBD
BSA receipt and acceptance of Delivery of End Item Documentation Package	TBD

Payment terms are Net XX (*for large/foreign businesses enter 30, for small businesses enter 10*).

Invoices shall be directed to BSA's Accounts Payable Department via electronic or regular mail.

Electronic invoice submittal is preferred by BSA. The Contractor shall make every effort to have systems in place that allow for electronic invoice transmittal. Invoices shall be submitted in PDF format, via email, to apinvoices@bnl.gov.

Hard copy invoices, if applicable, shall be directed to BSA's Accounts Payable Department, Contracts Division, Bldg. No. 400D.

The Contractor shall indicate the final invoice by clearly marking such invoice as "FINAL." A copy of the final invoice must be submitted to BSA's Contractual Representative.

- C. **Freight:** Freight terms shall be (*for North American shipments insert:*) FOB BSA National Laboratory, 98 Rochester Street, Upton, Long Island, New York.

D. **Labelling, Packing, and Notification Requirements:**

More detailed instructions are available at <https://www.bnl.gov/shipping/> or by emailing WarehouseOperations@bnl.gov.

Labeling Requirements

All shipments must comply with the following labelling requirements. Failure to do so may result in delayed payments. Labels must be placed on the upper left portion of the container when possible and contain:

Brookhaven National Laboratory
Brookhaven's Contract Number
98 Rochester St.
Upton, NY 11973
Box Number / Number of Boxes

Packing List Requirements

Every shipping container must include a packing list. The packing list must be clearly identified and affixed to a side of the shipping container. At a minimum, the packing list should contain the following information:

- Ship from company name and address
- Supplier contact name
- Supplier telephone number
- Ship to address as instructed by Buyer/Contracts Specialist
- Brookhaven's Contract number
- Part number, description, quantity, unit of measure, and corresponding BNL Contract line number

Special Notification of Delivery

If a shipment is heavier than 3,000 lbs., longer than 8 feet in any direction, or has a high center of gravity, it will require rigging upon receipt and the shipper shall provide a minimum of 3 days' advance notice of delivery to BNL's Receiving Department via email to WarehouseOperations@bnl.gov. The shipper must obtain acknowledgement from BNL that delivery notification was received. Failure to contact BNL will result in delays during delivery and any resulting demurrage will be at the shipper's expense.

Marking of Outer Package

Each shipping container (transport package) with a gross weight equal to or greater than 400 lbs. (180 kilograms) must be marked with the center of gravity location and gross weight on at least one side or end panel. It must also meet the requirements of the latest revision of [ASTM Standard D 3951](#), Standard Practice for Commercial Packaging. This applies to any item shipped (originally packaged or repackaged) to BNL or other locations cited in the Contract.

Protection of Material and Equipment

The Contractor shall employ procedures which ensure adequate protection of material and equipment during shipment and while in storage. Such protection shall include special environmental packaging as necessary. All items shipped (originally packaged or repackaged) to BNL or other locations cited in the PO or Contract shall comply with the requirements for preservation, packaging, and marking as stated in the latest revision of [ASTM Standard D 3951-10](#), Standard Practice for Commercial Packaging.

Unacceptable Packing Material

Do not use vermiculite as a packaging material inside any shipments to BNL unless required by shipping regulation. "Vermiculite" as defined in this specification means the hydrous, silicate mineral classified as a phyllosilicate that has been expanded by heating. Vermiculite used as a packaging material in shipments must be determined to be free of asbestos (<1% chrysolite, amosite, crocidolite, tremolite, anthophyllite, and actinolite). All shipments containing vermiculite must be accompanied by documentation stating that the vermiculite used is certified asbestos-free. This documentation must be affixed to the outside of the package.

Ocean Shipments

Foreign shipments transported via ocean freight must use the [10+2 REQUIREMENTS Form](#) (PDF). The form must be completed by the Contractor 72 hours before departure of the vessel. Send form to the following:

Interfreight Harmonized Logistics, Inc.

Fax: 516-371-6880

bnl@interfreight.net

Brookhaven National Laboratory

WarehouseOperations@bnl.gov

Once the form is submitted to customs, an acceptance number will be provided. Freight will not be loaded onto a vessel without an acceptance number. Failure to submit the 10+2 form may result in fines. It is the Contractor's responsibility to make sure the 10+2 form is submitted 72 hours prior to the vessel sailing. BSA shall withhold payment from the Contractor's final invoice for any penalties incurred by BSA due to late or incorrect ISF filing by the Contractor.

IV. AUTHORIZED REPRESENTATIVES

- A. **BSA's Technical Representative:** TBD, telephone no. TBD, TBD (email), is BSA's Technical Representative, and shall act as liaison between BSA and the Contractor in technical matters only.
- B. **BSA's Contractual Representative:** TBD, telephone no. TBD, TBD (email), is BSA's Contractual Representative. Any change or modification in the terms and conditions of this Contract shall require the written approval of BSA's Contractual Representative or designee.
- C. **Contractor's Technical Representative:** TBD, telephone no. TBD, TBD (email), is the Contractor's authorized Technical Representative, and shall act as point of contact between BSA and the Contractor in technical matters only.
- D. **Contractor's Contractual Representative:** TBD, telephone no. TBD, TBD (email), is the Contractor's authorized Contractual Representative and shall act as liaison between BSA and the Contractor.

V. PATENT CERTIFICATION

Upon Contract expiration and completion of all work authorized under this Contract, the Contractor shall complete and execute Attachment D "Patent Certification" clearance form and forward it to the Contractual Representative. Failure to do so may result in the withholding of final payment or a portion thereof.

VI. GOVERNMENT PROPERTY *Reserved*

VII. ADDITIONAL TERMS

The provisions of Brookhaven Science Associates, LLC, General Terms and Conditions for Commercial Items and Services, *(Rev 23, May 2025)* are incorporated herein and made a part hereof.

The provisions of Brookhaven Science Associates, LLC, General Terms and Conditions for Non-Commercial Items, *(Rev.23, May 2025)* are incorporated herein and made a part hereof.

The provisions of Brookhaven Science Associates, LLC, Supplemental Terms and Conditions for Work by Contractors Onsite at Brookhaven National Laboratory, *(Rev.23, May 2025)* are incorporated herein and made a part hereof. In accordance with Article 8 of these Supplemental Terms and Conditions, the Contractor is required to provide a Phase Hazard Analysis (PHA).

This Contract is issued pursuant to Brookhaven Science Associates General and/or Supplemental Terms and Conditions listed above, which are incorporated herein by reference. The complete text of these terms and conditions can be viewed at [PPM | Terms and Conditions](#). Should the Contractor have any questions and/or not have access to these terms and conditions, contact the Buyer or Contracts Specialist immediately.

The following documents attached hereto and made a part hereof, contain additional provisions of this Contract:

Attachment A Statement of Work, 2/27/2026

Attachment B Specification No. SPC-UTQMD-CFN-001, 2/5/2026

Attachment C BNL QA 101

Attachment D Patent Certification Form

This Contract does not bind nor purport to bind the Government of the United States.

ACCEPTED:

CONTRACTOR

**BROOKHAVEN SCIENCE
ASSOCIATES, LLC.**

By: _____

By: **TBD** _____

Name: _____

Name: **Mark Brock** _____

Title: _____

Title: **Subcontracts Supervisor** _____

Date: _____

Date: **See Electronic Signature** _____

The Contractor shall sign and return one (1) copy of this contract via email to the attention of Mark Brock, Procurement and Property Management Division at mbrock @bnl.gov.

STATEMENT OF WORK (SOW)
for
Ultralow Temperature Quantum Materials/Devices measurement system
(UTQMD)

Date

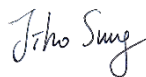
Prepared by:



2/27/2026

Aaron Stein, Technical Representative
Center for Functional Nanomaterials

Approvers:



02/18/2026

Jiho Sung, Staff Scientist, Center for Functional
Nanomaterials;

Quality Assurance

Nicole Magliulo

2/23/2026

Nicole Magliulo
Quality Management Office

ESH Coordinator



02/19/2026

Wai-Lin Ng
Center for Functional Nanomaterials

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"This document contains BNL procurement requirements specifications for solicitation and technical evaluation. The information is high-level and does not include the details necessary for "development" or "production" as defined under 15CFR § 772.1, 22 CFR §120.33, or 10 CFR Part 810; therefore, it does not constitute export-controlled technology or technical data."

1. SCOPE

This Statement of Work (SOW) will be the basis for the procurement of an ultralow temperature quantum materials/devices measurement system (UTQMD) for the Center for Functional Nanomaterials (CFN) & the Co-design Center for Quantum Advantage (C2QA) at Brookhaven National Laboratory. The UTQMD instrument, hereafter referred to as “Instrument” shall be designed as a cryostat with an integrated high-field magnet, providing an ultralow base temperature. The system uses a helium-4 pulse tube to precool to 4K and a helium-3/helium-4 mixture to reach <10 mK temperature. The low temperature helps to minimize thermal fluctuations and thus improve the quantum coherence of the qubits/enable to observe quantum phenomena such as superconductor, exciton condensate physics. It shall support (1) electronic transport and optical spectroscopy of two-dimensional heterostructure devices and (2) RF dielectric-loss characterization using high-Q microwave resonators for superconducting qubit materials – both at temperatures < 10 mK, with expected cooling power of (typical 14 μ W) on the experimental flange at 20 mK. The UTQMD should be a cryogen-free (“dry”) dilution refrigerator system and should include two separate vacuum and radiation shield set to support these two measurements. As an instrument offered to general users in a NSRC, the Instrument shall be engineered for safe, user-friendly operation (hardware and control software) and demonstrate a proven track record of low-maintenance, low-downtime operation. This document outlines the Contractor’s responsibilities and obligations necessary for completing the requirements as set forth herein.

1.1 Background

BNL’s mission and vision emphasizes integrating strengths in materials, nanoscale science, photon science, and computer science capabilities and expertise to drive the information science revolution in Quantum Information Science and Technology (QIST). Consistent with the Lab Plan’s “Future Quantum Materials” QIST thrust, CFN is developing state-of-the-art instrumentation for the user community, and the UTQMD is expected to create strong synergies with the Quantum Materials Press (QPress) by enabling feedback-driven cycles of materials discovery, device fabrication and low-temperature characterization. This effort will be carried out in partnership with the Co-design Center for Quantum Advantage (C2QA) to align measurement capabilities with quantum information device performance needs. The installation of UTQMD will strengthen BNL’s leadership in quantum materials discovery and characterization.

1.2 Definitions/Acronyms

ASME	American Society of Mechanical Engineers	JHA	Job Hazard Analysis
ATP	Acceptance Test Procedure	MRB	Material Review Board
BNL	Brookhaven National Laboratory	MRP	Material Requirements Planning
BSA	Brookhaven Science Associates	MXC	Mixing Chamber
C2QA	Co-design Center for Quantum Advantage	NFPA	National Fire Protection Association
CFN	Center for Functional Nanomaterials	NRTL	Nationally Recognized Test Laboratory
DC	Direct Current	QA	Quality Assurance
DR	Dilution Refrigerator	QIST	Quantum Information Science and Technology
ESH	Environment, Safety, and Health	QPress	Quantum Materials Press
FAT	Factory Acceptance Test	RF	Radio Frequency
GHS	Gas Handling Unit	SOW	Statement of Work
GUI	Graphic User Interface	UTQMD	Ultralow Temperature Quantum Materials/Devices Measurement System

2 APPLICABLE DOCUMENTS

In the event of a conflict between the Technical Specification and the SOW, the Contractor shall immediately notify the BSA Contractual Representative who shall in each instance determine which document takes precedence and advise the Contractor accordingly. Failure to notify BSA of a document conflict shall not relieve the Contractor's responsibility to ensure full compliance to all requirements.

The following documents are an integral part of the SOW; the applicable revision level will be the latest that is in effect at the time of award:

Document Number	Document Title
10CFR851	DOE Worker Safety and Health Program
BNL QA-101	Supplier Quality Assurance Requirements
ASME B30.26	Rigging Hardware
ANSI Z136.1	Safe Use of Lasers
ASME BPV	Boiler and Pressure Vessel Code
ASME B31.3	Process Piping -2015 Edition
NFPA 70	National Electrical Code – 2020 Edition http://www.nfpa.org/aboutthecodes/AboutTheCodes.asp?DocNum=70
NFPA 70E	Standard for Electrical Safety in the Workplace – 2021 Edition or the most current version in place
ASMEY14.5M-2008	Dimensioning and Tolerancing
JHA	BNL Job Hazard Analysis Template (Attached)

3 REQUIREMENTS

In accordance with the applicable specifications, referenced documents, and instructions as defined in this Statement of Work, the contractor shall be responsible for the engineering, purchase of all materials, manufacturing, assembly, inspecting, testing, delivering, installing, commissioning and training on the Instrument.

Safety and code compliance must be paramount at every stage of a project, from the initial concept to the final installation and use. Electrical design and installation must follow the National Electrical Code NFPA 70 and Standard for Electrical Safety in the Workplace NFPA 70E.

3.1 Contractor Responsibilities

3.1.1 Technical Performance

In accordance with SOW/Technical Specification document and all referenced documents, the Contractor shall engineer, manufacture and assemble the Instrument. The Contractor shall first test the Instrument at the Contractor's facility, then deliver, test and commission the Instrument at BNL. It is the Contractor's responsibility to ensure that the Instrument complies with all the specification requirements and the safety codes and standards referenced herein.

3.1.2 Tooling/Fixtures/Test Equipment

The Contractor shall be responsible for supplying all materials, including (but not limited to) the construction of all tools, jigs, fixtures, and test equipment required to complete the engineering, manufacturing, assembly, inspecting, testing, delivering, and installing the Instrument. Gases, cryogenics, and samples for testing will be supplied by BNL.

3.1.3 Factory Acceptance Testing

The Instrument shall be tested at the Contractor's facility to demonstrate full compliance with the SOW/Technical Specification. The Instrument shall be fully assembled and be representative of the final as-delivered configuration for Final Acceptance Test (FAT). The tests listed in the SOW/Technical Specification shall be conducted on the Instrument at the Contractor's facility according to the approved Acceptance Test Procedure (ATP), prior to delivery to BNL.

All test data shall be documented in the Factory Acceptance Test report and sent to BSA prior to the Instrument delivery. Acceptance of the test data is contingent upon data review and written approval by BSA technical representative prior to shipment of the Instrument.

3.1.4 Installation

3.1.4.1. The Contractor shall provide a footprint of the Instrument and utilities requirements.

3.1.4.2. The Contractor shall be responsible for all Instrument installation. BSA will not supply personnel for installation tasks other than identified herein. Unloading of containers and trucks, and staging of the Equipment at the installation site, shall be performed by BSA riggers. The Contractor shall not be permitted to drive BSA-owned forklifts for any reason. The Contractor shall submit an installation work plan that includes all onsite installation activities in a step-by-step format at least two months before the scheduled start of installation activities. This plan is a necessary prerequisite for obtaining a work permit and must be approved by BSA prior to the start of installation activities. The person (or persons) responsible for all onsite installation activities shall participate in the work planning process for the purpose of obtaining a work permit. This shall be completed during the Technical Review as well as after the Technical Review in video conferencing and/or teleconferences.

3.1.4.3. All training and safety requirements identified during the work planning process (or by BSA ES&H/Training staff members) shall be successfully completed by Contractor personnel who are participating in onsite installation work prior to issuance of a work permit and prior to installation.

3.1.4.4. BSA shall be responsible for offloading and rigging of heavy equipment. Contractors may operate their own rigging/cranes that are installed as part of their equipment, original equipment manufacturer (OEM) components. These must first be inspected by BSA.

3.1.4.5. Electrical tools/equipment needed for installation shall be inspected by BSA prior to use at BNL, to ensure compliance with electrical safety standards referenced herein.

3.1.4.6. The Installation Work Plan shall include at a minimum:

- a. Work instructions/ installation plan: job steps or detailed step-by-step instructions
- b. Prerequisites: preparatory actions to be performed before work can be started and any administrative and physical requirements (for example, review/ inspection/ approval of equipment, training, etc.)
- c. Precautions: Identify hazards and precautions that must be observed during performance of work in the Job Hazard Analysis (JHA)
- d. Identification for hold points and other monitoring
- e. Operational limits imposed
- f. Coordination of tasks
- g. Special conditions for working alone
- h. Scheduling, coordination, and notification
- i. Changes impacting configuration management
- j. ES&H work controls for mitigating identified hazards based on the following hierarchy, as practical:
 - i) Elimination/substitution of hazards
 - ii) Engineering controls
 - iii) Administrative controls and work practices
 - iv) Personal protective equipment

3.1.4.7. Note that disposal of waste packing material shall be the responsibility of BSA.

3.1.4.8. The Contractor shall supply all bolt-on wheels, transportation carts and custom tooling required for the installation of the Instrument which will remain the property of the Contractor at Contract Completion.

3.1.4.9. During assembly of all components, proper alignment and tightening shall be assured by the use of appropriate tools (e.g., torque wrenches) and by following the procedures recommended by the manufacturer, including the use of alignment tools and bolt-tightening sequences, etc.

3.1.4.10. Installation includes hardware and software setups for computer-based control of the Instrument.

3.1.5 **Acceptance Testing at BNL**

The Instrument shall be tested at BNL by the Contractor to demonstrate full compliance with this SOW and the Technical Specification documents. The tests listed in the SOW and the Technical Specification documents shall be conducted on the Instrument at BNL according to the ATP, prior to final acceptance.

3.1.6 **Commissioning**

The Contractor shall be responsible for the commissioning of the Instrument. Commissioning shall be defined as all test and demonstration work performed on site at BNL. Commissioning of the Instrument on site in its final configuration shall be performed with samples supplied by BSA and

agreed to by Contractor. The Instrument shall be able to demonstrate the performance specifications as outlined in the separate Technical Specification document.

3.1.7 Training

During the commissioning phase of the contract, the Contractor shall provide hands-on training at BNL for two (2) BSA scientific and/or technical staffs for the safe operation and maintenance of the Instrument. This training shall be conducted at BNL for a period of three (3) working days during commissioning for the Instrument.

3.2 Design and Design Review Requirements

3.2.1 Preliminary Design Report

A preliminary design to meet all the technical requirements shall be presented to BNL staff. A three-dimensional depiction in .easm file, CAD file, or similar format, shall be provided to BNL. A video conference meeting shall be held to discuss the design and facilitate questions. Changes to be incorporated shall be agreed upon by both BNL and the Contractor.

3.2.2 Final Design Report

A final design that incorporates and documents changes from the preliminary design shall be submitted to BNL. The final design shall be a three-dimensional depiction in .easm file, CAD file, or similar format. The instrument design in an .easm file or similar format shall be submitted. A video conference may be convened if deemed necessary.

3.2.3 Approval to Proceed

BSA must approve in writing the final design, including any modifications introduced at the Final Design Review Meeting before the Contractor proceeds with ordering material and components.

3.3 Manufacturing Requirements

3.3.1 Acceptance Test Procedure

The contractor shall provide an ATP that is in accordance with this SOW/Technical Specification and all reference documents noted herein. The ATP must be approved by BSA prior to use. The ATP shall ensure that all test requirements as defined in the safety codes referenced herein are achieved.

3.3.2 Commissioning Reports

A Commissioning report shall be provided by the Contractor. The report shall include, but not limited to, the results of all tests performed including graphs, printouts, drawing updates, calibration data, and control system tuning parameters. The Contractor shall be responsible for remedying any noncompliance regarding the requirements in this SOW/Technical Specification, subject to BSA approval. A summary sheet including demonstrated remedy of any non-compliances shall be included. Final acceptance is contingent on BSA data review and written approval by BSA.

3.4 Management

3.4.1 Program Plan

Within the timeframe listed in Section 5, the Contractor must deliver a detailed program plan which includes a milestone schedule defining the design validation, manufacturing, inspection, testing, installation and commissioning phases in sufficient detail to allow regularly scheduled progress monitoring. It shall be submitted for review and approval by BSA within the timeframe indicated in the Contract.

3.4.2 Progress Teleconference

Beginning within the timeframe listed in Section 5, monthly program technical and progress teleconferences and/or meetings between the Contractor and BSA shall be held at a mutually scheduled time. The discussions shall include the Contractor's progress, technical and contractual questions, presentations of analysis or testing results, design reviews, value engineering, trouble shooting, material status, tooling status, resources, and manufacturing issues.

3.4.3 Manufacturing/Inspection/Test Plan

A manufacturing plan shall be developed, which identifies the manufacturing processes of work required to complete all efforts as identified in this SOW. It shall be submitted for review and approval by BSA within six (6) weeks after award of contract. The plan may be a single document, or may make use of existing travelers or other suitable planning and control documents. It shall address the Material Requirements Planning (MRP), and at a minimum include manufacturing, inspection, and test steps including identification of critical manufacturing operations and parts/subassemblies showing integrated flow into the end item(s). If the Contractor does not have an MRP system in place, then details for purchasing all raw materials, shop loading (by work center), and a schedule for manufacturing all components and assemblies through packaging and shipping shall be provided by the Contractor.

3.4.4 Performance Reporting

Throughout the life of the contract, the Contractor shall supply a written report by the fifth of every month to the BSA technical representative and contractual representative clearly detailing progress through the prior month with respect to the program. This shall include milestone charts, detailed progress, open items, problems, recommended solutions, and risk mitigation.

3.5 Documentation and Data Management

3.5.1 All the documentation provided by the Contractor shall be in English.

3.5.2 All engineering drawings shall:

- a. Be formatted in accordance with ASME Y14.5M-2008
- b. Have views shown in third angle projection
- c. Preferably use metric dimensions and tolerances
- d. Have a separate parts list
- e. Have part number designations in accordance with LT-ENG-RSI-STD-002

3.5.3 The Contractor shall provide an electronic copy (email, plus either CD ROM or USB data storage device) of all required documentation as noted in this SOW. Manufacturing drawing CAD files of all components and assemblies shall be in any of these formats: STP, SAT, IGS, IAM, or IPT.

3.5.4 End Item Documentation Package

The Contractor shall provide an electronic copy of the End Item documentation package in accordance with the schedule set forth in this SOW. The End Item documentation package shall consist of the following documentation:

- a. Acceptance Test Reports and copies of test certificates
- b. All manuals
- c. Completed travelers (documents following instrument throughout manufacturing process)
- d. Manufacturing/ Inspection/Test data for inspections performed at incoming inspection or during manufacturing
- e. Certificate of Conformance as specified in Section 4
- f. Copies of specification/ data sheets for viewports.
- g. Copies of the Contractor's (or Subcontractors') weld certification/test reports
- h. Documentation of required proof test procedures and results for vacuum vessels
- i. Certification that all required pressure relieving devices used are approved by either the ASME or National Board of Boiler Inspectors (NBBI) Acceptance Test Reports and copies of test certificates

3.5.4.1 Documents

The Contractor shall provide an electronic copy (on a USB stick or BSA identified FTP site, in Microsoft Office format, or drawing format described in section 3.4 with formal "controlled copies" in PDF format) of the following documentation:

- a. All documentation identified in this SOW
- b. All parts and assembly solid models
- c. User/operational manual

3.5.4.2 Operation and Maintenance Manuals

The Contractor shall provide the operation and maintenance manual in electronic format with formal PDF format on a USB stick and provide two (2) hardcopies.

Operation and maintenance manuals shall include:

- 3.5.5 A title to the front cover, identifying the title of the project and the date of issue
- 3.5.6 A table of contents
- 3.5.7 A general description of the Instrument installation
- 3.5.8 Copies of any permits to use copyrighted material
- 3.5.9 A technical description of each system and subsystem to allow the correct and safe operation, maintenance and repair of all systems and subsystems installed, including:
 - i. *Operating manuals at all modes of operation described in Technical Specification document*
 - ii. *Standard maintenance operations not limited to beam alignment and adjustment of lens operations*
 - iii. *Trouble shooting techniques*
 - iv. *Recommendations for preventive maintenance*
 - v. *Disassembly, repair, cleaning and adjustment procedures*
 - vi. *List of recommended spare parts to keep on hand*

vii. *Software documentation and source code for open-source processing routines*

3.6 **Configuration Management**

The Contractor shall establish and maintain a configuration control system to assure that all end items (including spares) are of the proper configuration, and that all approved configuration changes are incorporated at the specified effectivity points. Records shall be maintained verifying the configuration of each item.

3.7 **Software Requirements**

The Contractor shall provide and maintain the capability to receive and send CAD files of all components or assemblies in STP, SAT, IGES, IAM, or IPT file format at the request of BSA.

3.8 **Environment, Safety and Health (ESH)**

The Contractor shall design and build all equipment in accordance with ESH requirements in the appropriate specifications, codes, standards, and other documents listed in Section 2 of this SOW.

3.8.1 **Contractor Work on Site at BNL**

BSA will identify and designate space necessary to safely perform staging and installation activities. Any cordons, stanchions, or barriers required to designate and isolate the Contractor's work area will be supplied by BSA.

Contractor shall submit a Job Hazard Analysis (JHA) identifying work hazards associated with the installation of the instrument including a list of hazardous materials and Safety Data Sheets (SDS) to BSA for approval with the Installation Work Plan.

3.8.2 **Electrical Safety**

3.8.2.1 All electrical equipment and components must be manufactured in accordance to U.S. standards and comply with NFPA70. The instrument shall be listed by a Nationally Recognized Testing Laboratory ([NRTL](#)). If the instrument is not NRTL-listed, the Contractor shall arrange for a third-party NRTL Field Evaluation to be done onsite at the completion of installation. The cost of the field evaluation shall be included in the quote. Any additional charges require BNL's prior written approval. Note that CE is not on the list of OSHA approved NRTLs. A report on the results of the field evaluation must be provided to BNL's Electrical Authority Having Jurisdiction (AHJ), and approved before the equipment is to be energized.

3.8.2.2 The contractor shall supply the following to BSA:

3.8.2.2.1 Documentation for the equipment installation, operation, service and maintenance that describes operation, shutdown, safety concerns, and nonstandard installations

3.8.2.2.2 Schematics, drawings, and bill of materials describing power feeds, voltages, currents and parts used for construction, maintenance, and operation of the equipment.

- 3.8.2.2.3 Safety Requirements and emergency shutdown procedures of equipment including lockout/tagout (LOTO) requirements.
 - 3.8.2.2.4 Documentation of specific hazards associated with the equipment.
 - 3.8.2.2.5 Documentation shall be provided if special tools, PPE, or other equipment is necessary for proper maintenance and operation of equipment.
- 3.8.2.3 The contractor shall include safety-related design concepts for electrical equipment and installations. Following methods can be considered:
- 3.8.2.3.1 Installing finger-safe components, covers, and insulating barriers to reduce access to exposed energized electrical conductors and circuit part due to inadvertent movement
 - 3.8.2.3.2 Substituting ≥ 50 volts control circuit with non-hazardous < 50 volt circuits.
 - 3.8.2.3.3 Separate enclosures for hazardous and non-hazardous electrical conductors and circuit parts.
- 3.8.2.4 The Contractor shall ensure that the equipment is marked with the NRTL certification or Field Evaluation label to include the manufacturer, model #, manufacturing date, drawing number, current draw, power, frequency voltages entering and leaving the control cabinets, and equipment. Caution, Warning, or Danger labels shall be affixed to the exterior describing specific hazards and safety concerns. Refer to ANSI Z535, Series of Standards for Safety Signs and Tags, for more information on precautionary labeling of electrical system or equipment
- 3.8.2.5 All exposed electrical connections/terminations ≥ 50 volts shall be covered so that workers are protected from the hazards of electrical shock during operation and when performing service & maintenance. Plexiglas® is not allowed; Lexan® may be used for inadvertent contact. Electrical insulation must be Low Smoke, Zero Halogen (LSZH), Fire Retardant, and rated for its application. Non-LSZH cables shall be permitted in length < 50 feet.
- 3.8.2.6 The contractor shall include safety-related design concepts for electrical equipment and installations. Following methods can be considered:
- 3.8.2.6.1 Installing finger-safe components, covers, and insulating barriers to reduce access to exposed energized electrical conductors and circuit part due to inadvertent movement.
 - 3.8.2.6.2 Substituting ≥ 50 volts control circuit with non-hazardous < 50 volt circuits.
 - 3.8.2.6.3 Separate enclosures for hazardous and non-hazardous electrical conductors and circuit parts.
- 3.8.2.7 Electrical wiring must be installed in a neat and workmanlike manner.

3.8.2.8 Teflon® shall not be used as an insulation material, except in small quantities

3.8.3 Pressure Safety

For process piping requirements, the contractor shall use ASME B31.3 or equivalent approved by BSA in writing. The liquid nitrogen, gaseous nitrogen, and cooling water lines shall be designed to ASME B31.3 requirements. In all cases, 100% proof testing shall be completed on each component to 150% of the maximum working pressure for hydrostatic tests, or 110% of the maximum working pressure for pneumatic tests.

To prevent inadvertent overpressurization of vacuum vessel(s), the system must be supplied with burst disk per ASME BPV Code.

3.8.4 Fire Safety

3.8.4.1 The Contractor shall ensure that any electrical equipment or wiring minimizes combustibility by the use of flame-resistant materials.

3.8.4.2 The Contractor shall ensure that any cable tray minimizes the possibility to damage cables. All cables to be installed in a cable tray shall be rated for Cable Tray use. The cable shall have the TC (Tray Cable) and NRTL markings. If a Contractor elects to use a cable not marked for Tray Cable use, then the Contractor shall use a cable that will meet IEEE 1202 or equivalent approved by BSA in writing. BSA shall approve the use of all non-NRTL marked tray cables in advance of the Contractor ordering the cable.

3.8.4.3 All other cables shall be Low Smoke Zero Halogen (LSZH). Non-LSZH cables shall be permitted in length <50 feet.

3.8.5 Attachment Points and Lifting Fixture

3.8.5.1 The Contractor shall provide lifting eyes, either permanent or removable, on each piece of equipment or stand of sufficient capacity and number to allow the equipment to be safely lifted.

3.8.5.2 All lifting fixtures must meet the requirements specified in ASME B30.26 (Rigging Hardware) and ASME BTH-1 (Design of Below-the-Hook Lifting Devices, Design Category "A").

3.8.6 Radio Frequency Safety

Any component using radio frequencies (i.e. plasma sources), should be labeled with the appropriate warnings following 29 CFR 1910.97 - ANSI C95.2-1982. Once installed radiofrequency radiation exposures limits shall not exceed Threshold Limit Values (ACGIH TLV – 2016) and Safety Levels (ANSI/ IEEE C95.1).

3.9 Packaging

In preparation for shipping, the Instrument shall be covered or wrapped for protection against weather elements, dust, and damage to projections, shock loads, impacts, and rubbing. The Instrument shall be

properly braced and cushioned within the packing so that it will not shift during handling and shipment. The method of shipping will be specified by BSA one month prior to delivery.

- 3.9.1 After the component or piece of the Instrument has been cleaned and tested, it shall be packed to ensure that it remains clean and free from damage during both storage and shipping. All seal faces and/or knife edges should be protected. Vacuum components shall be backfilled with dry nitrogen gas.
- 3.9.2 Added protection against contamination and humidity shall be given to the sensors and electrical connections. Durable materials, such as clean wood, bubble wrap, or similar non-contaminating materials, are to be used to support and protect the components from one another during transit and storage. Peanut-type material shall not be used for packaging.
- 3.9.3 All assemblies or sub-assemblies with *moving* parts shall be secured to prevent damage due to unwanted motion or vibration. The packaging must be designed to withstand acceleration loads in excess of twice the acceleration due to gravity ($>2g_0$).
- 3.9.4 Each packing crate shall be fitted with Shockwatch and Tipwatch sensors or an equivalent approved in writing by BSA.
- 3.9.5 The container shall be appropriately labeled with the listing of contents, destination, gross weight, location of center of gravity, purchase order number, and contact information.

3.10 Warranty

The Contractor shall provide a one-year warranty on all parts and labor commencing with BSA's acceptance of the installed and tested Instrument. A recommended spare parts list shall be available from the Contractor. Remote diagnosis and troubleshooting from Contractor shall be available upon request within 24 hours during regular workdays.

4 QUALITY ASSURANCE

The Contractor shall maintain and apply an effective QA program for the design, manufacture, and testing of all systems and equipment provided.

4.1 Quality Assurance Requirements

BSA Form BNL-QA-101, "Brookhaven National Laboratory Supplier Quality Assurance Requirements," forms a part of this statement of work. Specifically, this includes the following sections: 1, 2, 3, 3.1, 3.1.3, 3.1.4, 3.2 to 3.22, 4.0, 4.1, 4.2, 4.7, 4.7.1, 4.7.3, 4.10, 4.10.1, 4.16, 4.19, 4.21, 4.31, 4.34, 4.37, and 4.39.

4.2

Notification of Change to Design, Methods, or Processes

The Supplier shall immediately notify the Buyer of any significant changes (those that may affect form, fit, function, reliability, safety, or interchangeability) in product design, fabrication methods, materials, or processing from those used by the Supplier at time of Supplier's quotation or offer to the Buyer, which resulted in the PO.

4.3 **Records**

The Supplier shall retain objective evidence, including records, of the inspections and tests performed in the course of manufacturing, testing, inspecting, preserving, packaging, and preparation for shipment of procured items. These records shall be made available to the Buyer's representative for review upon request. These records shall be maintained for a minimum of three (3) years, unless otherwise specified in the procurement documentation, after the completion of the PO/contract.

4.4 **Franchised / Licensed "Distributor" Traceability**

Products that are not purchased directly from the Original Equipment Manufacturer (OEM)/Original Component Manufacturer (OCM) must be purchased only from a franchised/licensed distributor of the product being offered. The distributor shall ensure traceability of all products to the original equipment manufacturer by identifying the original manufacturer for each lot/date code on the Certificate of Conformance. The distributor shall provide a copy of the Manufacturer's certificate for the lot number being supplied, along with their franchised distributor certification. The distributor shall not use unapproved brokers (any company, person, or entity who is not an OEM/OCM) for the purchase of components and parts, unless pre-approval has been granted by Brookhaven Science Associates (BSA).

4.5 **Material Review Board (MRB) Authority: Reserved**

4.6 **QA Flow-Down**

The Contractor shall ensure a proper and complete flow-down of all applicable QA requirements to its suppliers and sub-tier suppliers.

4.7 **Certificate of Conformance**

With each shipment, the Contractor shall submit a Certificate of Conformance (C of C). In case of drop shipment, a copy of the certificate shall be submitted to BSA at the time of shipment. The certificate shall include the title of and be signed by an authorized representative of the company, and shall constitute a representation by the Contractor that:

A. Materials used are those which have been specified by BSA, and that the items delivered were produced from materials for which the Contractor has on file, reports of chemical or physical analysis, or any other equivalent evidence of conformance of such items to applicable specifications;

B. Processes used in the fabrication of items delivered were in compliance with applicable specifications included as part of the PO/contract, or BSA-approved procedures or specifications;

C. The items as delivered comply with all applicable drawings, specifications, deviations/waivers, and other requirements of the procurement documentation; and-

D. When specified, cleaning and cleanliness requirements have been completely satisfied. The C of C shall reference the Contractor's applicable cleaning procedures.

5 DELIVERABLES

The Contractor shall supply the following:

Item	Deliverable	Due	BSA Approval
1	Program Plan	2 nd week after award of contract	Yes
2	Technical and Progress Teleconference	2 nd week after award of contract/monthly	No
3	Performance reports	5 th of each month after receipt of Program Plan and within 2 weeks of key project milestones	No
4	Preliminary Technical Design	4 th week after contract award	Yes
5	Final Technical Design	5 th week after contract award	Yes
6	Equipment footprint and laboratory requirements	6 th week after contract award	Yes
7	Manufacturing/Inspection/Test plan	6 th week after contract award	Yes
8	Factory acceptance testing	50 th week after award of contract	Yes
9	Installation Work Plan, JHA	52 nd week after award of contract	Yes
10	Delivery of the Instrument	52 nd week after award of contract	Yes
11	Site Acceptance Test and Commissioning Report	56 th week after award of contract	Yes

6 SCHEDULE/ MILESTONES

The following schedules are for planning purposes only. Schedule requirements set forth in the contract will take precedence.

Milestone	Calendar weeks after award
Preliminary Technical Design Submitted	4 th week
Final Technical Design Approved	5 th week
Factory Acceptance Test Report	50 th week
Installation Work Plan, JHA	52 nd week
Delivery to BNL	52 nd week

Commissioning of instrument	55 th week
Training on instrument	55 th week
Delivery of End Item Documentation Package	55 th week

7 VALUE ENGINEERING

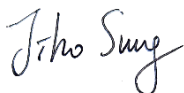
The Contractor is encouraged to make recommendations for changes that might lead to an improvement in performance, reliability, quality, safety, or reduction in cost. Simplicity in operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production. Where it appears a substantial improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified, it is desirable that their use be investigated.

Ultralow Temperature Quantum Materials/Devices measurement
system Technical Specification

QA Category: A-2

CENTER FOR FUNCTIONAL
NANOMATERIALS
BROOKHAVEN NATIONAL
LABORATORY
BROOKHAVEN SCIENCE ASSOCIATES
UPTON, LONG ISLAND, N.Y. 11973

PROJECT: Ultralow Temperature Quantum Materials/Devices measurement system Technical Specification		
Version 1	SPC-UTQMD-CFN-001	Feb. 05, 2026
	Document Number	Date



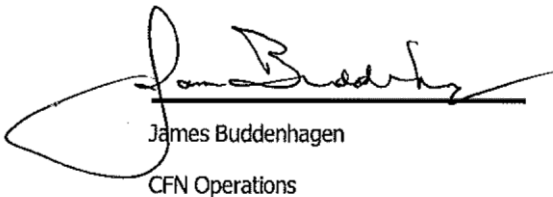
Jiho Sung
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VERSION CONTROL SHEET

VERSION	DESCRIPTION	DATE	AUTHOR	APPROVED BY
1	First Release	Feb. 05, 2026	J. Sung	See cover sheet.

“This document contains BNL procurement requirements and specifications for solicitation and technical evaluation. The information is high-level and does not include the details necessary for "development" or "production" as defined under 15 CFR § 772.1, 22 CFR § 120.33, or 10 CFR Part 810; therefore, it does not constitute export-controlled technology of technical data.”

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PROJECT:

**Ultralow Temperature Quantum Materials/Device measurement system
Technical Specification**

Version 1	SPC-UTQMD-CFN-001	Feb. 05, 2026
	Document Number	Date

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1. INTRODUCTION

1.1 Overview

The ultralow temperature quantum materials/devices measurement system (“UTQMD”) is a cryogen-free dilution-refrigerator platform with an integrated high-field magnet, designed for electronic transport measurements and optical spectroscopy of two-dimensional heterostructure devices, as well as RF dielectric-loss characterization using high-Q microwave resonators for superconducting qubit materials. The system offers an automated bottom loading fast sample exchange, enabling cooldown from room temperature to 20 mK temperatures within nine (9) hours. It is equipped with multiple DC and RF wiring lines and provides optical access via ports and fiber feedthroughs.

The UTQMD at the Center for Functional Nanomaterials shall be aimed at a milli-Kevin characterization platform for a broad range of quantum materials and devices, including graphene, transition metal dichalcogenides, two-dimensional magnets, and superconducting qubit materials.

1.2 Definitions

1.2.1 Acronyms

BNL	Brookhaven National Laboratory
CFN	Center for Functional Nanomaterials
ES&H	Environmental Safety and Health
UTQMD	Ultralow Temperature Quantum Materials/Devices measurement system
GSH	Gas Handling System
FSE	Fast Sample Exchange
SMA	Sub Miniature version A
FC/APC	Ferrule Connector/Angled Physical Contact
NRTL	Nationally Recognized Testing Laboratory

	Version 1	SPC-UTQMD-CFN-001	Feb. 05, 2026
		Document Number	Date

2. SCOPE OF SUPPLY

2.1 General Information

This document describes the required specifications for the ultralow temperature quantum materials/devices measurement system (“UTQMD”) to be installed at CFN. The system shall be designed as a cryostat with an integrated high-field magnet, providing a ultralow base temperature. The system uses a helium-4 pulse tube to precool to 4K and a helium-3/helium-4 mixture to reach <10 mK temperature. The low temperature helps to minimize thermal fluctuations and thus improve the quantum coherence of the qubits/enable to observe quantum phenomena such as superconductor, exciton condensate physics. It shall support (1) electronic transport and optical spectroscopy of two-dimensional heterostructure devices and (2) RF dielectric-loss characterization using high-Q microwave resonators for superconducting qubit materials – both at temperatures < 10 mK, with expected cooling power of (typical 14 μ W) on the experimental flange at 20 mK. All electrical components must be NRTL certificated. As an instrument offered to general users in a NSRC, the system shall be engineered for safe, user-friendly operation (hardware and control software) and demonstrate a proven track record of low-maintenance, low-downtime operation.

3. Configuration & performance requirements

This section defines the required features and describes the performance specifications of the system. The system should be a fully integrated instrument with subsystems that include the pulse tube cryocooler & its helium compressor, cryostat and radiation shields, gas handling system, automated bottom loading fast sample exchange, DC & RF wiring, optical access ports, cryogen-free magnet, control and diagnostics, graphical user interface (GUI) software, and all cables, hoses, and pumps for operation.

The UTQMD system shall deliver a technical performance that is equivalent or superior to the following LD250 dilution refrigerator system offered by Bluefors Oy:

3.1 Pulse tube cryocooler and compressor:

- Cryomech pulse tube cryocooler with a remote motor provides the cryogenic cooling power of the dilution refrigerator measurement system without the need for liquid cryogens.
- Precools components of the cryostat insert to ~50K and ~3K through integrated heat switches.
- Cools the radiation shields of the cryostat during operation
- Provides two pre-cooling stages for the circulated helium-3 gas as it returns to the cryostat
- Provides additional cooling power to heat sink experimental wiring
- The compressor for the helium mixture circulation is not of membrane type (but scroll type) and thus, there is no risk of rupture, which could lead to helium mixture loss or contamination
- Pulse tube cryocooler can be easily replaced without removing the dilution unit or any of the experimental plates for ease of maintenance.

- Cryomech model PT425-RM cryocooler with expected cooling powers at 1st stage (50W at 45K) and at 2nd stage (2.35W at 4.2K).
- Voltage specification: 440/480 VAC 3Ph 60Hz
- Remote motor and ballast tanks in a separate shelf for vibration mitigation

3.2 Cryostat and radiation shields:

- Provided with 3-section vacuum can, 50K shield, 4K shield, and still shield compatible with cryogen-free 14 T magnet and fast sample exchange. system without the need for liquid cryogens.
- Separate 2-section vacuum can, 50K shield, 4K shield, still shield, mixing chamber shield, and RT magnetic shield for superconducting qubit experiments without magnet and fast sample exchange.
- Dilution unit can be easily replaced without removing any of the experimental plates for ease of maintenance
- Fully welded main heat exchanger, not prone to developing leaks (like soft solder joints) and thus provides indefinite trouble-free operation
- Modular system design allows prompt replacement of parts in case of part failure
- Gas gap heat switches used as a part of the support structure of the cryostat insert enables clear access to the experimental flanges
- Heavy duty support frame with added machined aluminum top plate (~200kg) for mounting the cryostat for vibration isolation
- Expected cooling power of typical 14 μ W on the experimental flange at 20 mK; more than 250 μ W on the experimental flange at 100mK; more than 360 μ W on the experimental flange at 120mK

3.3 Gas Handling System (GHS):

- Fully automatic GHS that manages the helium gas circulation in the system
- Core unit contains control electronics, pressure gauges, helium gas tanks, and vacuum and pneumatic components
- Booster unit contains the circulation boosters for the main circulation circuit
- Dedicated pumps for the helium circulation system are the main circulation pump and circulation boosters. The main circulation pump is a multistage roots pump. It is maintenance free with 6 year average service life with advanced diagnostics The circulation boosters are high-vacuum turbo pumps. The circulation boosters are located in GHS booster unit instead of on top of the cryostat. This eliminates the electrical noise and mechanical vibration transfer and gives better access to the top plate services
- Extra service booster (Pfeiffer HiPace80 vacuum turbo pump) for pumping vacuum can, fast sample exchange loading system and/or service manifold
- The sealed pumps and compressors in Gas Handling Systems are oil-free
- The cryostat is electrically isolated from the pumping system with electrically insulating spacers, clamps, and centering rings inserted into a break in each pumping line close to GHS
- The cryostat is mechanically isolated from GHS booster unit by an edge-welded metallic T-damper

- Long-life cold trap to condense all the possible contaminant gases in the helium circulation and prevent clogging, allows to use the system for 3 years without interruption and without need for external liquid nitrogen trap
- Control software with full system diagnostics. Instrument cart includes temperature controller, 4K heater power supply, and display for high-performance industrial PC
- Industrial PC offers fully independent and versatile operation of the dilution refrigerator measurement system, and software is pre-installed, configured, and tested for optimal performance and control of the system
- Resistive heater (40W, 10 Ohm/ 2A max) at the 4K flange for faster warm-up, which is integrated with system control software
- Voltage specification for GHS core unit: 115V 60Hz (Single phase power)

3.4 Control and diagnostics:

- The system is controlled with programmable logic controller (PLC) system. It provides application programming interface (API) with OPC-UA protocol to all supported devices and enables access control to the listed parameters of devices
- The Control software can be used for remote control and automatic cooldown and warm-up processes. The control software installation package includes the software and all necessary drivers
- Even when not connected to the computer, the system safely recovers and restores from a power failure. Following the restoration of power after a power failure, the system attempts to restart the circulation by opening the necessary valves and starting the main circulation pump. The rebooting time of PLC system is 3-5 min. In any event, regardless of the length of time that the system has been without power and whether the system is still cold enough to restart circulation, the system is in a safe mode following a loss of power. If there is a long power failure, the mixture naturally returns to the helium tanks through the system of overpressure valves.
- Temperature controller and 4K heater power supply feature 8-channel resistive temperature measurement, four heater controls operational in manual or closed loop mode, intuitive device touchscreen and web interface and application programming interface (API) enabled via ethernet connection.

3.5 Bottom loading fast sample exchange (FSE) system:

- Fully automatic load lock system for sample loading and unloading, which enables the sample exchange without warm-up of the whole cryogenic measurement system and thus reduces the total turn-around time of the system (cooldown time to <20 mK may exceed 7 hours but will typically be less than 9 hours)
- <20 mK base temperature and 50 μ W cooling power at 100 mK
- Load lock mechanism with bellows and gate-valve to deliver robust, leakproof operation. No o-ring based sliding seals that are prone to leak
- Grounding wiring with detach mechanism that protects sample while it is loaded into the system

- The detach mechanism automatically disconnects the physical contact between the FSE insert tube and the FSE insert still stage, which breaks the thermal connection between these two temperature stages. This also disconnects the grounding wiring from the FSE load lock breakout
- FSE insert diameter of 58 mm, inner diameter of sample space, 53mm, and 25mm diameter hole in the center of the probe head for top optical access
- Two micro D-sub 25pin connectors and the 4 RF connectors for DC/RF lines
- 12 Phosphor-bronze twisted pairs (24 wires) from probe room temperature break-out box to the probe detachment mechanism
- Probe thermometry package that includes additional dedicated temperature controller with thermometer, RuOx (0.015 ~ 100 K range)
- Attocube nanopositioner-ready package includes mounting brackets (gold plated copper) for nanopositioner stack inside FSE probe, and adapter wiring inside FSE sample space for plugging in the nanopositioners

3.6 Experimental wiring:

- DC installation set includes room temperature break-out box with 3 fischer connectors
- Wiring for positioners and scanners from room temperature to mixing chamber includes 6 low-ohmic wires for positioners (<5ohm) and 18 (Cu+PhBr) wires for scanners and measurements.
- 12 twisted pair experimental wiring (American wire gauge (AWG) copper) from room temperature connector box (24-pin FISCHER) to 4K flange. 12 twisted pair experimental wiring (NbTi/CuNi) from 4K flange to the mixing chamber flange. Intermediate 25-pin micro-D breakout at the 4K flange. Termination with 25-pin micro-D breakout at the mixing chamber flange. EM shielding for 12 twisted wire pairs
- RF installation set KF63 12 SMA wires from room temperature to mixing chamber. Vacuum KF63 flange with 12x hermetic SMA feedthrough plus 1x aluminum and 1x gold thermal anchoring flanges with F/F SMA bulkheads for 50K and 4K stages. Gold plated copper thermal anchoring flanges with F/F SMA bulkheads for Still, and mixing chamber.
- At least 8 Semi-rigid coaxial lines operating up to 18 GHz. Out of the 8 lines, 6 should be 0.86mm SCuNi-CuNi from room temperature to mixing chamber with low-loss infrared filter, and 2 should be 2.19mm SCuNi-CuNi from room temperature to 4K and 0.86mm NbTi-NbTi. These 2 wires will be used as readout lines and 6 SCuNi-CuNi only lines will be used as RF drive lines. At least 4 cryogenic low noise amplifier equivalent to LNF-LNC4_8F (4-8 GHz) installed.
- Wiring for powering 4 low noise amplifier (LNA); Power block array for LNA; Power supply for 4 LNA

3.7 Cryogen-Free magnet:

- Rated field (z axis) of 14 Tesla at 4.2K
- 76 mm bore (~68mm sample space)
- Compensation coil is integrated within the magnet structure above the main coil to decrease the stray fields at the low temperature stage of the system

- Persistent mode switch (includes persistent switch harness) increases the stability of the superconducting magnets over long periods of time. The expected field drift with the persistent mode switch is <100 ppm/h
- Field homogeneity: +/- 0.1% over 1cm DSV (diameter of spherical volume)
- Integrated Cu/HTS (high temperature superconductor) 120 Ampere current leads
- 4-quadrant power supply system that includes power supply, power supply programmer, all cabling and magnet current-lead quick connection rail

3.8 Optical access ports:

- Top vertical optical access that includes clamping set with blinds at room temperature, 50K, 4K and still. UV antireflection coated fused silica for window material
- 3 inches “bossed” optical port with blind flanges for horizontal optical access
- Optical fiber installation set (FC/APC Hermetic feedthrough + optical fibers) comprising room temperature breakout box, thermalization flanges at 50K, 4K, Still, Cold plate and mixing chamber stages, and mixing chamber breakout. At least 4 optical fibers will be included. These should be polarization maintaining fibers with model PM1550, PM980, PM-780-HP and PM630HP.
- Optical fiber installation set for bottom loader FSE, FC/APC Hermetic feedthrough with PM780HP fiber

3.9 Microwave Readout Module:

- Microwave wide band near quantum limited readout module with high signal to noise ratio.
- This module features gain bandwidth 5 GHz to 7 GHz, gain in 80% of band 11 dB, added noise in 95% of band <918 mK, gain ripple within 200 MHz window < 5 dB, and signal to noise ratio improvement 9 dB expected
- This module includes travelling wave parametric amplifier (TWPA), parametric amplifier cold-finger assembly with dual magnetic shield, filtered DC lines for TWPA flux bias control, low-noise high-electron-mobility transistor (HEMT) amplifier and filtering/isolation microwave components

3.10 Service feature, system installation and commissioning:

- Technical support teams are ready to help with any issues regarding the functionality and operation of the system. Support is provided by expert technical personnel who are experienced with all the products and response is provided with 1 business day via email
- System packing, shipping, installation, and commissioning
- Lab layout design: 3D system installation layout in accordance with the information about the facility space provided by the customer
- 24 months warranty for products, 12 months warranty for third-party products, and 6 months warranty for spare parts.

4. Optional Items

4.1 System care service

Express access to original equipment manufacturer spare parts, which are shipped out in 2 business days and delivered with a call-off sales process to ensure fast delivery, saving time to resolve problems. Using original equipment manufacturer spare parts ensures optimal performance level of systems.

Spare parts are provided from stock in US production facility

5. Other requirements

5.1 Magnetic fringe field

The vendor should provide a drawing on the magnetic fringe field when the cryogen-free magnet operating at full power, with 5-Gauss lines clearly marked.

5.2 Delivery and Installation

The complete system shall be delivered on site within 53 weeks upon the order being placed. The vendor must provide a guide for CFN to prepare the lab facility for the system and arrange an engineer to come on site for installation after all parts are delivered defect-free. The Contractor shall be responsible for installation of all components and provision of all interconnecting power, communication, control and instrument cabling, and all liquid and gas piping, and tubing between components and to the supply points identified by BNL. Contractor shall be responsible for providing all materials and supplies required for installation and for verification of specified performance. All installation work shall be in accordance with applicable National Codes.

BROOKHAVEN NATIONAL LABORATORY

SUPPLIER QUALITY ASSURANCE REQUIREMENTS

(BNL-QA-101)

PO/Contract No.: 476517

INSTRUCTIONS: At least one sub-clause in Clause 3.1 must be selected which will automatically invoke Clauses 3.2 through 3.22 collectively on purchase orders. If applicable, the Special Requirements of Section 4.0 need to be individually selected and can be modified as required.

1.0 PURPOSE & SCOPE

- 1.1 This document establishes quality assurance requirements to which Suppliers to Brookhaven Science Associates (BSA) must conform when specified in the procurement documentation.
- 1.2 This document contains two main sections. Section 3.0 covers the general requirements that are applicable to all Suppliers. Section 4.0 contains special quality requirements that are applicable only when specifically invoked in the procurement documentation.

2.0 DEFINITIONS

- 2.1 The term Procurement documentation means the purchase order (PO), contract, subcontract, Request for Proposal (RFP), Request for Quotation (RFQ) or other written agreement with the Supplier (seller) in which the requirements of BSA are incorporated.
- 2.2 The term Buyer means BSA operating Brookhaven National Laboratory, acting by and through its Procurement & Property Management Division (PPM) issuing the PO/contract.
- 2.3 The term Supplier (seller) means the legal entity, which is the contracting party, with the Buyer with respect to the procurement documentation.
- 2.4 The term article or item means a product and/or a service.

3.0 GENERAL REQUIREMENTS

Unless otherwise specified in the procurement documentation, the following General Requirements apply:

3.1 Supplier's Quality System and Quality Requirements

The Supplier must have and maintain an effective quality system that will, as a minimum, comply with all of the requirements as designated by the following:

☐ **3.1.1** A quality system certified/registered to the ISO 9001 standard: (latest revision as of the date of issuing the procurement documentation).

☐ **3.1.2** A quality system that meets the requirements of the ISO 9001 standard: "Quality Management Systems – Requirements" (latest revision as of the date of issuing the procurement documentation).

☒ **3.1.3** Conformance to Supplier's/Manufacturer's quality program or system.

☒ **3.1.4** Other: Refer to procurement documentation, (e.g. PO, Statement of Work [SOW], specifications, drawings) for quality requirements.

NOTE: Clauses 3.2 through 3.22 apply to all POs and will be included collectively in other procurement documentation when required/specified.

3.2 Assessment by Buyer

The Supplier's quality system is subject to assessments by the Buyer's Representative(s) for conformance with the requirements of the PO. Supplier or Distributor must allow BSA representatives, BSA customers, and regulatory agencies right of entry into the Supplier's facilities to determine and verify product, processes, records, personnel, material, procedures, and systems.

3.3 Change Approval

No change(s) are permitted to be made to any Buyer requirements, (e.g. part number, model number, etc.) without the prior written approval of the Buyer.

3.4 Responsibility for Subcontractors

It is the responsibility of the Supplier to impose applicable requirements from this document upon their subcontractors. Additionally, the Buyer reserves the right to disapprove, in writing, any subcontractor. For nuclear facility requirements per NQA-1, the Supplier must incorporate NQA-1 requirements in sub-tier procurement documents.

3.5 Responsibility for Conformance

The Supplier is responsible to provide items that conform to the requirements of the PO regardless of any assessments, surveillances, inspections and/or tests by the Buyer or its representatives at either the Supplier's or Buyer's facility. The Buyer reserves the right to request failure analysis and corrective action for non-conforming articles or items submitted or supplied to the Buyer. The Supplier is responsible for notifying the Buyer of any recalls or alerts associated with this PO.

When NQA-1 requirements are imposed, prior to shipment, the Supplier must report any nonconformance that cannot be corrected to full conformance and obtain approval from BNL for possible Use-As-Is or Repair dispositions. Additionally, for commercial grade services, a service performance record must be provided that documents the adequacy of the service performed to the identified critical characteristics documented in the PO/contract.

3.6 Protection of Material and Equipment

The Supplier must employ procedures that assure adequate protection of material and equipment during shipment and while in storage. Such protection must include special environmental packaging, as necessary. All items shipped (originally packaged or repackaged) to BNL or other locations cited in the PO or contract, must comply with the requirements for preservation, packaging and marking as stated in the latest revision of ASTM Standard D 3951 Standard Practice for Commercial Packaging.

3.7 Measuring and Test Equipment (M&TE) Calibration

The Supplier must calibrate any M&TE used in the fulfillment of the PO requirements against certified standards that are traceable to the National Institute of Standards and Technology (NIST), or some other recognized national or international standard, or physical constant. The Supplier must notify the Buyer of any condition found during the calibration, servicing or

repair of measuring and test equipment that can affect the end item requirements.

3.8 Suspect Counterfeit Parts

- The Supplier shall verify the procurement source and associated documentation certifying authenticity.
- Appropriate incoming inspection and / or test methods shall be used to detect potential suspect counterfeit items.
- The Supplier shall flow this requirement down to all sub-tier suppliers to prevent the inadvertent use of suspect counterfeit items.
- Distributors shall not modify, rework or repair any items shipped on this order.
- For reference only, a list of authorized/franchised distributors for electronic components can be found at: <https://www.trustedparts.com/>

* For more information refer to the following Department of Energy website: <https://www.energy.gov/ehss/corporate-reporting-analysis/databases/suspectcounterfeit-and-defective-items>

3.9 Electrostatic Discharge Control

Items that are susceptible/sensitive to electrostatic discharge (ESDS) must be handled and packaged to protect them from damage. Items and/or packages must be labeled to indicate the susceptibility to electrostatic discharge.

3.10 Electrical or Fire Protection Equipment, Material, and Systems

All electrical or fire protection equipment, material, and systems delivered to BNL must be certified, listed, or labeled by a Nationally Recognized Testing Laboratory (NRTL). The CE mark is NOT a recognized NRTL certification mark. (For a listing of OSHA-recognized NRTLs, refer to <http://www.osha.gov/>)

For electrical or fire protection equipment, material, and systems which no NRTL accepts, certifies, lists, labels, or otherwise determined to be safe, the Supplier must determine the equipment to be safe for its intended use. The determination must be made on the basis of test data. The determination and test data documents must be made available to BSA prior to or upon delivery for review and acceptance by the applicable BSA Authority Having Jurisdiction (AHJ).

In accordance with 29 CFR 1910.147(c)(2)(iii) whenever new machines or equipment are provided with energy isolating devices, those devices must be designed to accept a lockout device.

3.11 Hoisting & Rigging Equipment

All hoisting & rigging equipment used at BNL must meet the requirements of the latest applicable OSHA Regulations and ASME B30 Series standards for design, construction, markings, and proof load testing.

When proof load testing is required by the standards, a certificate must be provided upon delivery documenting the proof test.

3.12 Deleted – requirements were added to each Purchase Order as the “Advance Notification of Delivery.”

3.13 Powered Machine Shop Equipment

All powered machine shop equipment (e.g., lathe) delivered to BNL must meet the requirements of the latest applicable OSHA 1910 Regulations (e.g., part subpart O). Equipment purchased must include an integrated NFPA 79 compliant emergency stop and an anti-restart device.

3.14 Vehicle-Mounted Elevating and Rotating Aerial Devices

All vehicle-mounted and rotating aerial devices equipment used at BNL must meet the requirements of ANSI A92.2.

3.15 Self-propelled Elevating Work Platform Equipment

All self-propelled elevating work platform equipment (e.g., scissor lift) used at BNL must meet the requirements of ANSI A92.6.

3.16 Manually Propelled Elevating Aerial Platform Equipment

All manually propelled elevating aerial platform equipment used at BNL must meet the requirements of ANSI 92.3.

3.17 Boom Supported Elevating Work Platform Equipment

All boom supported elevating work platform equipment used at BNL must meet the requirements of ANSI 92.5.

3.18 Powered Industrial Trucks and Attachments

Powered industrial trucks (e.g. forklifts, hi-lows) must meet the requirements of the ANSI/ITSDF B56 series for design, construction, markings, and test loading. Industrial truck attachments must be approved by the truck manufacturer and supplied with an attachment data plate indicating the new truck capacities.

3.19 Used Industrial Equipment

For used industrial equipment (e.g. scissor lifts, forklifts, etc.), a Certificate of Conformance as defined in clause 4.16 must be provided as objective evidence and must additionally state that all maintenance and manufacturing alerts have been screened and all required repairs and improvements have been completed. The Supplier must provide records of the last year of maintenance.

3.20 Global Harmonized System Compliance

The supplier of chemicals must deliver the chemical in full compliance with the Department of Labor, Occupational Safety & Health Administration (OSHA)’s Globally Harmonized System (GHS) Hazard Communication Standard (29CFR1910.1200), available at:

<https://www.osha.gov/dsg/hazcom/HCSFinalRegTxt.html>

All hazardous chemicals delivered to BNL must be accompanied by an GHS Safety Data Sheets (SDS) with the format and content specified in 29CFR1910.1200. For information on the GHS SDS see OSHA Brief- Hazard Communication Standard: Safety Data Sheets Publication 3514 available at:

<https://www.osha.gov/Publications/OSHA3514.pdf>

All hazardous chemicals delivered to BNL after 12/01/2015 must have a label with the elements specified in 29CFR1910.1200 [product identifier; pictograms; signal words; hazard statement(s); precautionary statement(s); and manufacturer, importer, or distributor’s name, address, and telephone number]. For information on the GHS label, see OSHA Brief- Hazard Communication Standard: Labels and Pictograms Publication 3636 available at:

<https://www.osha.gov/Publications/OSHA3636.pdf>

3.21 Age/Shelf Life and Storage Control

The Supplier must have an effective storage and age control system for items where acceptability is limited by the age or manner of storage of the item. The system must include a method of identifying the expiration date on the containers in which material is delivered to the Buyer. Special handling conditions must be recorded on certifications and shipping documents covering the material delivered to the Buyer. At the time of receipt, the material must not have less than three-quarters of its shelf life remaining, without prior written approval from the Buyer for each shipment.

3.22 Product Recalls/Product Bulletins/Safety Alerts

Any and all product recall alerts, product bulletins, or safety alerts must be communicated by email directly to ProductRecallAlert@BNL.gov. Provide the Purchase Order Number(s) and names of purchasers with notification to assist BNL in locating and identifying the subject material.

4.0 SPECIAL REQUIREMENTS

The following Special Requirements are applicable only when specified in the procurement documentation or as indicated by check mark hereon. These Requirements can be modified as required.

INSTRUCTIONS: Since sub-clauses (e.g., 4.4.1) are tied to the main clause (e.g., 4.4), the requirements of the main clause will apply by default whenever any sub-clause is selected (regardless of whether the main clause was selected/checked).

☒ 4.1 Quality Assurance Program or Manual

The Supplier must submit a copy of their Quality Assurance Program or Manual with their proposal for review and evaluation.

☐ **Suppliers providing A1/A2 but without a quality system compliant with NQA-1, i.e., Commercial Grade Supplier,** must follow applicable Commercial Grade Dedication Instructions as defined in the SOW.

OR

☐ Supplier must utilize the QA Program audited and approved by BNL.

☒ 4.2 Configuration Control System

The Supplier must establish and maintain a system to assure that all end items (including spares) are of the proper configuration, and that all approved configuration changes are incorporated at the specified effectivity points. Records must be maintained to verify the configuration of each item.

☐ 4.3 Process Sheets, Travelers, etc.

The Supplier must maintain a system of process sheets, shop travelers, or equivalent means to define the sequence of manufacturing, inspection, installation and test activities to be performed. Flow sheets, or equivalent, must be provided for sign-off by designated inspection personnel at specified inspection and test points, including, as required, re-inspection and re-test points, to assure completion as well as proper sequencing of required operations.

☐ 4.4 Manufacturing/Inspection/Test Plan

Sixty (60) calendar days prior to performance of work, the Supplier must submit for the Buyer's approval a Manufacturing/Inspection/Test Plan for the item(s) to be produced. Once approved, changes/revisions must be approved by the Buyer prior to implementation. The Plan must satisfy one or more of the following as selected:

☐ **4.4.1** Identification of parts and subassemblies showing integrated flow into end item(s).

☐ **4.4.2** Identification of critical manufacturing operations, as well as inspection and test checkpoints.

☐ **4.4.3** The Plan may be a single document, or may make use of existing "travelers," or other suitable planning and control documents.

☐ 4.5 "Witness" Points

The Buyer reserves the right to designate selected manufacturing, inspection, and/or test operations as "witness" points. The Supplier must provide the Buyer with five (5) working days' notice in advance

of reaching such witness points during the manufacturing and test cycle of each item. For nuclear facility applications (NQA-1), witness points are required when Source Inspection is invoked. (see Clause 4.14)

☐ 4.6 Test and Inspection Procedures

Test and inspection procedures required to demonstrate satisfactory completion of requirements must be prepared by the Supplier and submitted to the Buyer for approval sixty (60) calendar days prior to use of such procedures. Once approved, changes/revisions must be approved in writing by the Buyer prior to implementation.

☒ 4.7 Special Processes

Processes (e.g., welding, brazing, bonding, plating, chemical machining, chemical coating, chemical cleaning, precision cleaning, heat treating, or waste processing) that either cannot be verified non-destructively or require a unique (special) non-destructive test/inspection (e.g., radiographic inspection, ultrasonic testing, pressure leak testing) must be performed in accordance with detailed written procedures. These procedures must specifically describe the exact manner in which the processes are to be performed. Additionally, the following requirements apply as selected:

☒ **4.7.1** Copies of special process procedures must be made available on request for review by the Buyer's representative.

☐ **4.7.2** At least sixty (60) calendar days prior to use on items deliverable to the Buyer, the Supplier must submit to the Buyer copies of all applicable process procedures for review and approval. Revisions or changes to Buyer-approved special process procedures must be submitted to the Buyer for review and approval prior to implementation.

☒ 4.7.3 Qualification of Procedures, Facilities, Equipment and Personnel

The Supplier must, prior to use, qualify the procedures/specifications, facilities, equipment and personnel that will be used for the performance of special processes. Only those personnel who have been qualified to perform a specific special process must be used to perform that process. Records of such qualification must be available to the Buyer's representative upon request.

4.8 Qualification of Procedures, Facilities, Equipment

Superseded by Sub-clause 4.7.3

4.9 Qualification of Special Process Personnel

Superseded by Sub-clause 4.7.3

☒ 4.10 End-Item Documentation Package

The Supplier must provide a documentation package for each shipment of the item(s) supplied, which consists of objective evidence of compliance with PO requirements. This documentation package must be complete, legible, indexed, and traceable to the item supplied. Additionally, the following requirements apply as selected:

☒ **4.10.1** Copies of reports of all required or necessary inspections, examinations and tests, properly validated by the Supplier's authorized personnel.

☐ **4.10.2** A listing of the as-built configuration of each delivered item; this may be defined by the use of drawing numbers and revisions, unique parts lists or other such means of positive identification.

☐ **4.10.3** Copies of nonconformance reports dispositioned as "rework/repair" or "use-as-is", and all BSA approved deviation/waivers.

- ☐ **4.10.4** Copies of material test certificates for specified materials, showing physical and chemical properties.

4.10.5 – Superseded by Clause 4.16

☐ **4.11 Release for Shipment**

The documentation package required in Clause 4.10 must be approved by the Buyer's representative prior to release of the item for shipment.

☐ **4.12 Shipment of Documentation Package to Buyer**

Three (3) copies of the documentation package required in Clause 4.10 must be shipped to the Buyer with or prior to each shipment of the purchased items.

☐ **4.13 Failure Reporting, Analysis and Corrective Action**

The Supplier must maintain a failure reporting, analysis and corrective action system that must, at a minimum, evaluate, analyze and correct failures occurring during qualification, first article and end-item acceptance testing and inspection. The results of all failure evaluations and analyses must be documented and available for review by the Buyer.

☐ **4.14 Source Inspection/Verification**

Items to be delivered require inspection, tests or surveillance by the Buyer's representative at the Supplier's facility. Five (5) working days advance notice, for acceptance inspections and tests, must be provided by the Supplier to the Buyer to permit scheduling of source inspection. For nuclear facility applications (NQA-1), source verification must be conducted in accordance with a Verification Plan to inspect, test, or examine at predetermined/Witness points (see Clause 4.5). Documented evidence of acceptance must be provided upon completion of all verifications.

☐ **4.15 Chemical and Physical Test Report**

One copy of the actual chemical and physical test report(s) for each heat, batch or lot must accompany each shipment. Test reports must list the actual parameters tested, the acceptable limits for each parameter, and must contain the actual readings taken during test.

☒ **4.16 Certificate of Conformance (C of C)**

With each shipment, per the procurement documentation, the Supplier must submit a Certificate of Conformance (C of C). In case of drop shipment, a copy of the certificate must be submitted to the Buyer at the time of shipment. The certificate must identify the purchased item/material/equipment by PO number and quantity, include the title of and be signed by an authorized representative of the company, and must constitute a representation by the Supplier that:

- A. Materials used are those which have been specified by the Buyer, and that the items delivered were produced from materials for which the Supplier has on file, reports of chemical or physical analysis, or any other equivalent evidence of conformance of such items to applicable specifications;
- B. Processes used in the fabrication of items delivered were in compliance with applicable specifications included as part of the PO/contract, or Buyer-approved procedures or specifications;
- C. The items as delivered comply with all applicable drawings, specifications, deviations/waivers and other requirements of the procurement documentation; and-
- D. When specified, cleaning and cleanliness requirements have been completely satisfied. The C of C must reference the Supplier's applicable cleaning procedures.

☐ **4.16.1 Additional C of C Requirements for Nuclear (NQA-1) Purchases**

A. The certificate must be signed by the person responsible for the quality function and whose function and position is described by the supplier's QA program/system.

B. The Supplier's Certification system must be described in their QA Program/system documentation and include the process for completing the C of C along with the review and Approval of the C of C prior to delivery.

C. The Buyer reserves the right to verify the validity of Supplier's certificates and the effectiveness of the Supplier's certification system. This verification will be commensurate with the Supplier's quality performance/history.

4.17 Report with Each Shipment

Superseded by Clause 4.10

☐ **4.18 First Article Acceptance**

Buyer acceptance of first article(s) is required prior to the production run. The first article(s) must be identified as such, including the PO number/contract, part number, and part name. The Supplier is required to:

- ☐ **4.18.1** Submit the first article(s) to the Buyer's representative for test/inspection to be conducted at the Supplier's facility by the Buyer's representative.
- ☐ **4.18.2** Submit the first article(s) to the Buyer for test / inspection by the Buyer at the Buyer's facility.
- ☐ **4.18.3** Submit the first article(s) to the Buyer together with documents showing data representing results of the Supplier's first article(s) test/inspection, including the actual dimension or value for each specified characteristic.
- ☐ **4.18.4** After Buyer acceptance of first article(s), all of the remaining units required by the PO/contract must be produced by the Supplier and the Supplier's suppliers using the same design, materials, processes, methods and tooling that were used to manufacture the approved first article(s). Any changes must have prior written approval from the Buyer.

☒ **4.19 Notification of Change to Design, Methods, or Processes**

The Supplier must immediately notify the Buyer of any significant changes (those that may affect form, fit, function, reliability, safety, or interchangeability) in product design, fabrication methods, materials, or processing from those used by the Supplier at time of Supplier's quotation or offer to the Buyer, which resulted in the PO.

4.20 Age/Shelf Life and Storage Control

Superseded by Clause 3.21

☒ **4.21 Serial Numbers**

The Supplier must assign/mark a separate and distinct serial number to each end-item in accordance with the procurement documentation. A record of the serial number, for each part number, must be maintained by the Supplier.

☐ **4.22 Lot or Batch Numbers**

For items furnished in accordance with the procurement documentation, the manufacturing lot or batch number must be indicated on the packing list, certifications and other applicable documents. Where impractical to mark individual parts due to size or

shape, the lot or batch number must be marked on identifying tags or the smallest unit package.

☐ **4.23 Material Traceability**

Materials used must be identified by material type, applicable specification and revision number, and be traceable to their lot and/or heat number(s). Traceability records must be available for review by the Buyer's representative.

☐ **4.24 Shipment Destination Other than BNL**

The material ordered is to be shipped to other than the Buyer's facilities. Copies of the data required in accordance with the procurement documentation must accompany the shipment; in addition, one copy of such data must be mailed to the Buyer on the same day that shipment is made.

4.25 Heat Treat Bars

Superseded by Clause 4.7

☐ **4.26 Burn-in**

Burn-in must be performed on each completed item, per the procurement specification or Supplier's Burn-In process approved by the Buyer. Records of burn-in testing, repairs and test results must be maintained and must be available to the Buyer's representative upon request.

4.27 Welding Procedures

Superseded by Clause 4.7

☐ **4.28 Weld/Braze Inspection Report**

A report(s) must be submitted that indicates the complete inspection of welds or brazes from the initial fit-up stage through final inspection. Inspection reports must be accompanied by all radiographic films, filler metal reports etc. The reports must contain the signature or stamp, and title of an authorized Supplier representative.

☐ **4.29 Radiographic Quality Requirements**

Items requiring radiographic inspection must be radiographed and processed in accordance with the Supplier's special process procedures that satisfy design specifications, standards or other procurement documentation requirements. Personnel reading and interpreting film must have been examined and certified. Responsibility for this certification must rest with the Supplier, whether the Supplier does the work or subcontracts to a specialized laboratory. A report of the findings must include the name of the reader and the signature and title of a responsible representative. The radiographic film and a reproducible copy of the report must accompany each shipment. An adequate method of identifying and cross-referencing each film exposure, report, and item must be provided. When parts are serialized, serial numbers must appear on the report and the film.

☐ **4.30 Nondestructive Test Reports**

All nondestructive testing must be conducted in compliance with the Supplier's special process procedures that satisfy the applicable provisions of the design specifications, or other procurement documentation requirements. Personnel and equipment utilized in performance of such tests must be qualified for the type of test performed. The Supplier must furnish with, or prior to, each shipment reports of such nondestructive examination of material or items furnished. These reports must be identifiable to the respective item or material including the specific section, joints or views of the item furnished. These reports must contain the signature and title of an authorized Supplier representative. When items are serialized, the serial numbers must appear on the reports.

☒ **4.31 Pressure or Leak Test Reports**

Test reports must be prepared for all pressure and leak tests. Such reports must state the requirement, the Supplier's test procedure number, and the observed result for each item, joint or connection tested. When items are serialized, the serial numbers must appear on the report. Reports must contain the signature/title of an authorized Supplier representative and must accompany each shipment.

4.32 Cleaning Certification

Superseded by Clause 4.16 D

☐ **4.33 Calibration Certification**

The Supplier must submit with each instrument/system a certification that the instrument/system has been calibrated and is ready for use. The certification must contain, at a minimum, the identity of the instrument/system, identification of the calibration procedure used, identification of the standards and/or equipment utilized for the calibration, and a statement that the calibration of the standards and/or equipment used is traceable to the NIST or some other recognized national or international standard, or physical constant. Unless otherwise specified, detailed support data must remain on file for minimum of three (3) years with the Supplier and must be available for review by the Buyer. The certification must also contain the signature and title of an authorized Supplier representative.

☐ **4.33.1** The Supplier will provide "As Found" (i.e., before) and "As Left" (i.e., after) measurements with the certification.

☐ **4.33.2 Additional NOA-1 Calibration Certification Requirements**

The certificate must be signed by the person responsible for the quality function and whose function and position is described by the supplier's QA program/system.

The Supplier's Certification system must be described in their QA Program/system documentation and include the process for completing the C of C along with the review and Approval of the C of C prior to delivery.

The Buyer reserves the right to verify the validity of Supplier's certificates and the effectiveness of the Supplier's certification system. This verification will be commensurate with the Supplier's quality performance/history.

The certificate must include the following:

- M&TE ID or Asset number
- The Supplier's calibration procedure
- The recommended frequency of calibration
- The calibration date
- The next calibration due date
- As-found calibration data must be reported in the certificate of calibration when calibrated items are found to be out-of-tolerance.
- The standards used for calibration
- The PO number
- Any additional technical and quality requirements, as necessary.

On the certificate, the supplier must also attest that the contracted calibration or test service has been performed in accordance with their ISO/IEC-17025 program, has been performed within their scope of accreditation, is traceable to NIST, the PO's requirements are met, and identify any and all nonconforming conditions.

4.34 Operating-Maintenance Manual

Documentation containing operating procedures, maintenance instructions, spare parts lists, and handling procedures must be submitted with the shipment of the first item.

4.35 Computer Software Configuration Management

The Supplier must have and maintain an effective software configuration management system. The Supplier's system must establish requirements for placing software under configuration control, provide for the positive identification of software, and the control of all software baseline changes.

- ☐ **4.35.1** The Supplier must submit a copy of their software configuration management procedure(s) with their proposal for review and evaluation, and specifications for software features, including requirements for safety, security, functions, and performance.

4.35.2 Superseded by Sub-clause 4.35.1

4.36 Computer Software Verification and Validation (V&V)

The Supplier must develop written procedures describing the controls applied to the design of software and the validation of the design through independent technical review. The procedures must provide for documentation of review activities, including requirements for documenting comments and resolution of comments. Supplier software designs and review documentation must be subject to review and approval by the Buyer.

The Supplier must test and verify computer software developed or modified to fulfill the requirements in the procurement documentation. The verification testing must be accomplished by a comparison of test results with those from other verified software, or by a comparison with results from analytical solutions or Buyer-approved alternatives.

4.37 Computer Software Notification of defects, new releases or other issues

For Safety Software, the Supplier must provide a process to (a) notify users of defects, new releases, or other issues that impact the operation and (b) mechanisms for the users of the software to report defects and request assistance in operating software.

4.38 Computer Software User Training

The Supplier must provide formal on-site, off-site or on-line training for users.

4.39 Records

The Supplier must retain objective evidence, including records, of the inspections and tests performed in the course of manufacturing, testing, inspecting, preserving, packaging, and preparation for shipment of procured items. These records must be made available to the Buyer's representative for review upon request. These records must be maintained for a minimum of three (3) years, unless otherwise specified in the procurement documentation, after the completion of the PO/contract.

4.40 Electrical, Fire Protection, or Scaffolding Equipment, Material, and Systems

Superseded by Clause 3.10

4.41 Hoisting & Rigging Equipment

Superseded by Clause 3.11

4.41.1 Powered Industrial Truck Attachments

Superseded by Clause 3.18. Supplier may not ship partial orders without prior written approval from BNL.

4.41.2 Custom-made Equipment

Superseded by Clause 3.11

4.41.3 Critical Lifts

Superseded by Clause 3.11

4.42 Marking of Outer Package and Hoisting & Rigging Services

Superseded by Clause 3.12

4.43 Franchised/Licensed "Distributor" Traceability

Products that are not purchased directly from the Original Equipment Manufacturer (OEM)/Original Component Manufacturer (OCM) must be purchased only from a franchised/licensed distributor of the product being offered. The distributor must provide documentation that the distributor is contractually authorized by the manufacturer as a franchised / licensed distributor along with the geographic region(s) covered by the authorization. The distributor must not use brokers (any company, person, or entity who is not an OEM/OCM or authorized by the OEM/OCM) for the purchase of items, unless pre-approval has been granted by Brookhaven Science Associates (BSA). For reference only, a list of authorized/franchised distributors for electronic components can be found at: <https://www.trustedparts.com/>

- ☐ **4.43.1** The distributor must ensure traceability of all products to the Original Equipment Manufacturer (OEM)/Original Component Manufacturer (OCM) by identifying the Original Equipment Manufacturer (OEM)/Original Component Manufacturer (OCM) for each lot and / or date code on the Certificate of Conformance. The distributor must provide a copy of the Manufacturer's certificate for the lot number being supplied, along with their franchised distributor documentation.

4.44 Power Machine Shop Equipment

Superseded by Clause 3.13

4.45 Aerial Lifts Equipment

Superseded by Clause 3.14

4.46 Self-propelled Elevating Work Platform Equipment

Superseded by Clause 3.15

4.47 Nuclear (NQA-1) Calibration Services

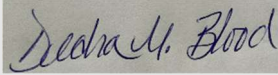
The service must be provided in accordance with the supplier's accredited ISO/IEC-17025 program and scope of accreditation. As-found calibration data must be reported in the certificate of calibration when calibrated items are found to be out-of-tolerance. The equipment and standards used to perform the calibration must be identified in the certificate of calibration. Subcontracting of these accredited services is prohibited. BNL must be notified of any condition that adversely impacts the supplier's ability to maintain the scope of accreditation. Performance of the services listed on this order is contingent on the Supplier's accreditation having been achieved through an on-site accreditation assessment by the accreditation body within the past 48 months. Any additional technical and quality requirements, as necessary.

4.48 Nuclear (NQA-1) Right of Access

BNL or designated representative may enter the premises of Supplier (or its sub-tier supplier) and review records for the purpose of inspection, surveillance, or quality assurance audit.

4.49 Nuclear (NQA-1) Partial Orders

Attachment D

PROCUREMENT AND PROPERTY MANAGEMENT DIVISION BROOKHAVEN NATIONAL LABORATORY Managed by Brookhaven Science Associates, LLC under contract to the U.S. Department of Energy ACQUISITION MANAGEMENT SYSTEM FORM	Form No. AMS-Form-032 Revision No. 2.0 Effective Date: September 23, 2024 APPROVED BY:  <u>Dedra Blood / 9/23/24</u>																				
Patent Certification																					
Contractor: _____ <i>Please check the appropriate box:</i> Contract Number: _____ ☐ Interim Certification or ☐ Final Certification Contractor hereby certifies that: 1. All procedures for identifying and disclosing subject inventions as required by the patent clause of the contract have been followed throughout the reporting period. 2. There were no subcontracts or purchase orders related to this contract involving research, development, and demonstration except as follows: _____ _____ 3. No inventions or discoveries were made or conceived in the course of or under this contract other than the following: <i>Please check the appropriate box:</i> Certification includes ☐ or does not include ☐ all subcontracts: <table border="1" style="width: 100%; border-collapse: collapse;"><thead><tr><th style="width: 20%;">TITLE</th><th style="width: 20%;">INVENTOR(S)</th><th style="width: 20%;">DATE REPORTED</th><th style="width: 20%;">iEDISON #</th><th style="width: 20%;">DOE "S" #</th></tr></thead><tbody><tr><td> </td><td> </td><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td><td> </td><td> </td></tr></tbody></table> <u>Attach additional sheet(s) if needed.</u> 4. The completion date of this contract is: _____ 5. The following period is covered by this certification: _____ Month Day Year To _____ Month Day Year 6. Contractor Signature: _____ Title: _____ Address: _____ Date of Certification: _____		TITLE	INVENTOR(S)	DATE REPORTED	iEDISON #	DOE "S" #															
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